Descriptive Analytics Report:

Analysis The Apple App Store At Scale

Exploring App Categories,Rattings,Pricing,Size,User Engagement,Languages & Device Support

Note on Data

This descriptive analytics report is based on an Apple App Store dataset containing information about **7,197 mobile applications**. The dataset provides a structured view of applications available through the App Store and contains information related to **application identification, application name, application size, currency, price, total user ratings, version-specific ratings, user rating scores, application version, content rating, application category, device support, screenshot availability, language support, and volume purchase program licensing**. These variables provide a useful foundation for performing descriptive analysis and understanding the characteristics and performance of applications across different categories.

The dataset contains **17 variables**, which together describe different aspects of an application. The **App ID** provides a unique identification value for each application, while the **Track Name** represents the name of the application. These fields allow individual applications to be identified and compared during the analysis. The dataset also contains **application size information in bytes**, which can be used to understand how large applications are and whether application size varies across different categories.

Another important part of the dataset is the **pricing information**. The **currency** field identifies the currency used for application pricing, while the **price** field represents the listed price of an application. The dataset contains both **free applications and paid applications**, making it possible to examine the distribution of applications according to pricing. This information can be used to sdominate the App Store, how paid applications distributed across categories, and how pricing differs between application types.

1. Objective

Build a descriptive analytics layer over Apple App Store applications to answer five recurring business questions:

1. **What categories contain the largest number of applications?**
2. **What is the distribution of free and paid applications across the App Store?**
3. **How are applications rated, and where is rating quality strongest or weakest?**
4. **Which applications and categories show the highest level of user engagement?**
5. **How do application price, size, and other characteristics vary across categories?**

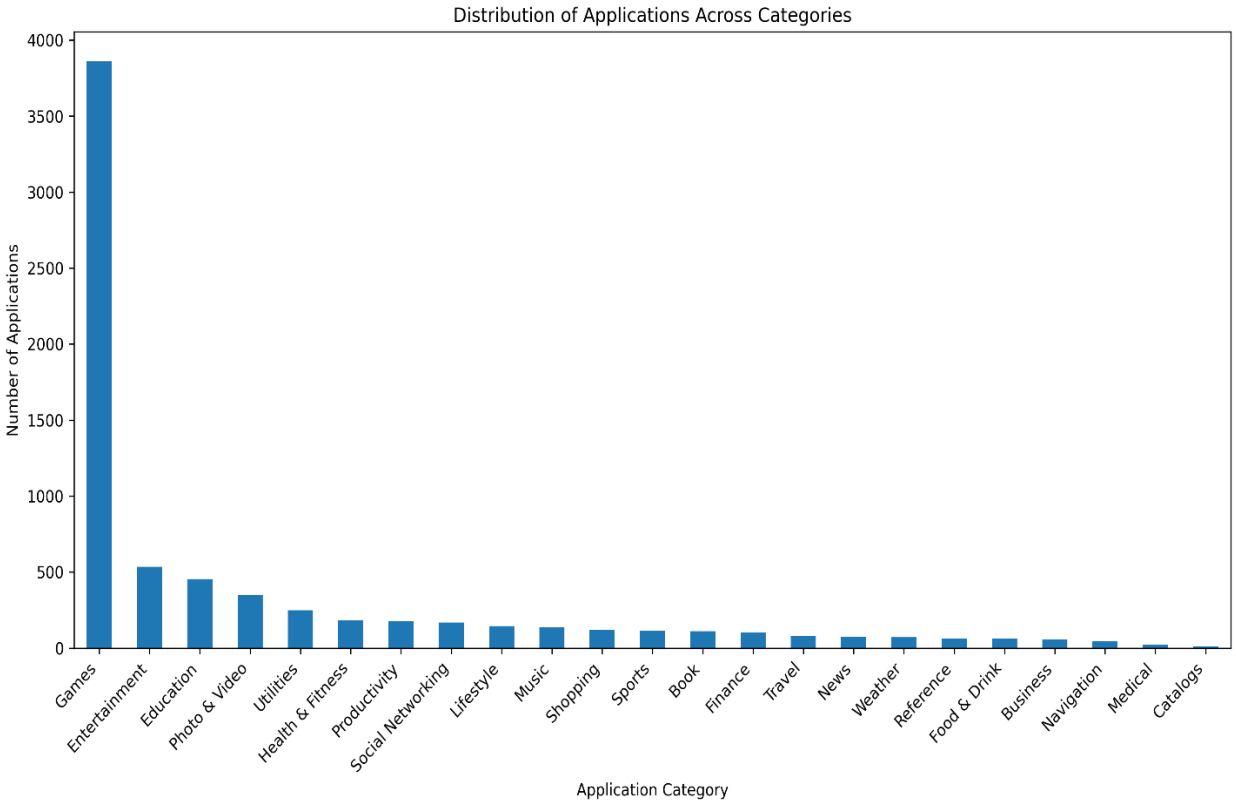
2. How This Would Work with Apple App Store Data

layer	purpose	Typical approach/tool
Data collection	Collect application information such as app name, category, price, ratings, size, language support and device support.	keggle
Data cleaning	Check missing values, duplicate records and inconsistent data before analysis.	Python/pandas
Data preparation	Organize and prepare the required application	Panda data frame

	variables for analysis and visualization.	
Data processing	Calculate application counts, category-wise values, pricing measures, rating measures and user-engagement measures.	Pandas
Data visualization	Create graphs to represent the patterns and comparisons found in the dataset.	Python/matplotlib

Report preparation	Present the analysis, graphs, interpretations, findings and conclusions in a structured report.	Microsoft word

3.application category distribution



- Interpretation:

The graph represents the distribution of applications across different categories in the Apple App Store dataset. The total dataset contains **7,197 applications** distributed across multiple application categories.

The **Games** category has the highest number of applications, with **3,862 applications**. The next highest category is **Entertainment**, which contains **535 applications**. **Education** has **453 applications**, while **Photo & Video** contains **349 applications**.

The **Utilities** category includes **248 applications**, followed by categories such as **Health & Fitness, Productivity, Social Networking, Lifestyle, and Music**, which have comparatively smaller numbers of applications.

At the lower end of the graph, **Catalogs** contains only **10 applications**. Therefore, the graph presents the numerical distribution of applications across all categories, with the number of applications varying considerably from one category to another.

• Insights:

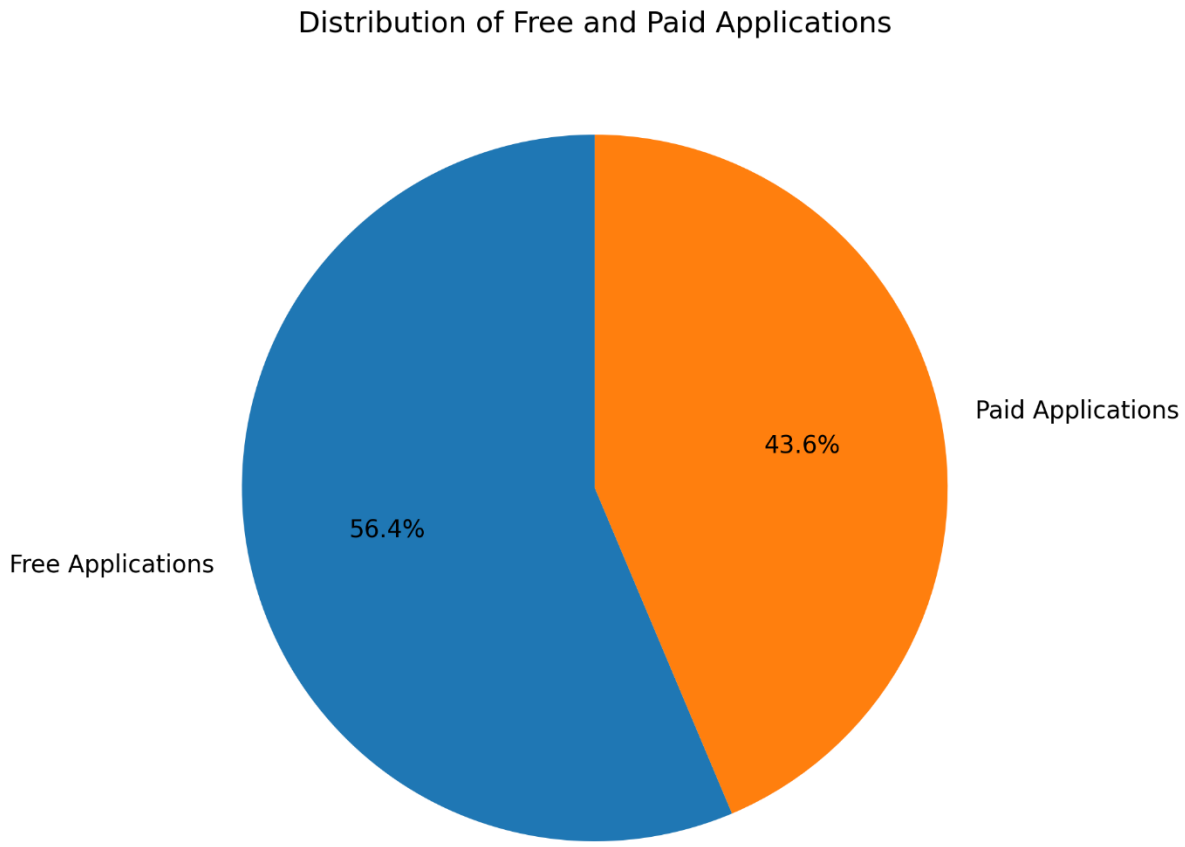
- Game is the most dominant category in the dataset, with **3,862 applications**.
- **Entertainment** is the second-largest category with **535 applications**, followed by **Education** with **453 applications**.
- **Photo & Video** and **Utilities** also have a noticeable number of applications, with **349** and **248** respectively.

- Categories such as **Health & Fitness, Productivity, Social Networking, Lifestyle, and Music** have comparatively lower application counts.
- **Catalogs** is the smallest category, with only **10 applications**.
- Overall, the graph shows a **highly uneven distribution**, with a very large concentration of applications in the games category.

KPI Defination used:

- **Total Applications – 7,197:** This shows the total number of applications included in the Apple App Store dataset.
- **Games – 3,862 applications:** Games is the largest category, showing that it has the highest number of applications compared with all other categories.

4. Free and Paid Application Distribution



- Interpretation:

The graph compares the number of **Free Applications** and **Paid Applications** in the Apple App Store dataset. The dataset contains a total of **7,197 applications**. Out of these, **4,056 applications are free**, whereas **3,141 applications are paid**.

The graph clearly shows that the number of free applications is higher than the number of paid

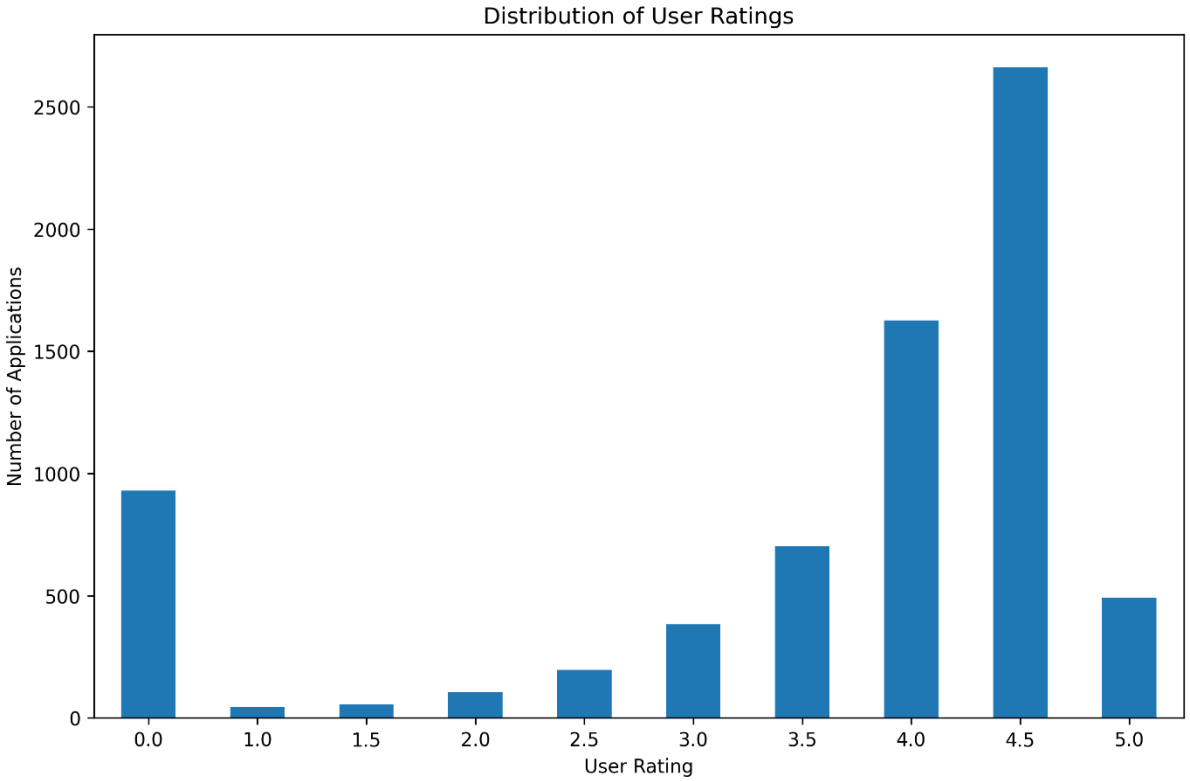
applications. Free applications represent approximately **56.36%** of the total applications, while paid applications represent approximately **43.64%**.

The difference between the two groups is **915 applications**, calculated from **4,056 – 3,141**. In other words, for every 100 applications in the dataset, approximately **56 are free and 44 are paid**. The graph therefore shows a noticeable but not extremely large difference between the two application type.

- ## Insights

- **Free applications** are significantly higher in number than paid applications.
- The graph clearly shows that **free applications form the majority** of the Apple App Store dataset.
- **Paid applications** represent a much smaller portion of the total applications.
- This indicates that the dataset is **strongly dominated by free applications**.
- Overall, there is a **large difference between free and paid applications** in the dataset.

5. Rating Distribution:



- **Interpretation:**

The graph represents the distribution of applications according to their user ratings in the Apple App Store dataset. The graph covers ratings from 0 to 5 stars and shows the number of applications available at each rating level.

The largest number of applications is found at a 4.5-star rating, with 2,663 applications. This is followed by the 4.0-star rating, which contains 1,626

applications. The 5.0-star rating has 492 applications, while the 3.5-star rating has 702 applications.

Among the lower rating categories, 3.0 stars contains 383 applications, while 2.5 stars contains 196 applications. The 2.0-star rating has 106 applications, and the 1.5-star rating has 56 applications. The 1.0-star rating contains 44 applications.

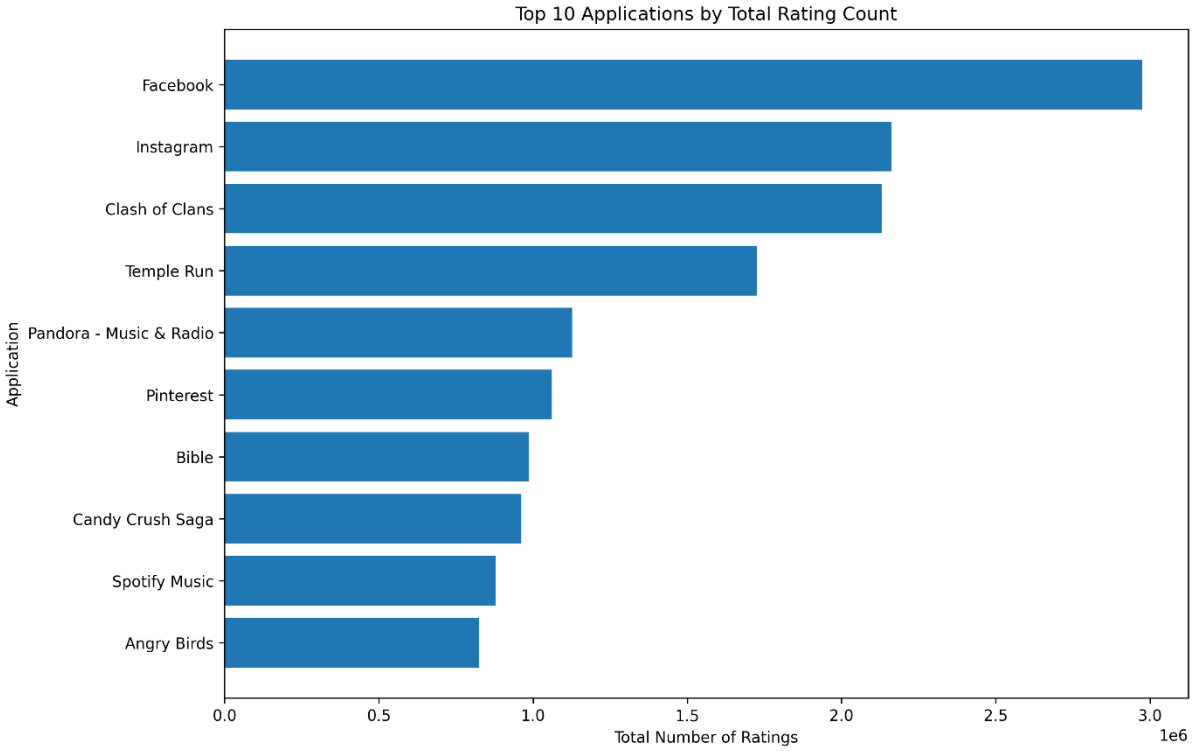
The graph also includes 929 applications with a rating value of 0, which is the second-largest group after the 4.5-star category. Overall, the graph displays application counts of 929, 44, 56, 106, 196, 383, 702, 1,626, 2,663, and 492 across the respective rating values from 0 to 5.

- ## Insights:

- The **4.0–4.5 rating range** contains a large number of applications.

- Applications with ratings around **4.0 and above** are more commonly represented in the dataset.
- Very low ratings such as **1.0 and 1.5** have comparatively fewer applications.
- Applications with a **5.0 rating** are present but are fewer than those in the main 4.0–4.5 range.
- Overall, the graph indicates that **most applications in the dataset have relatively positive user ratings**.

6. User Engagement



- **Interpretation:**
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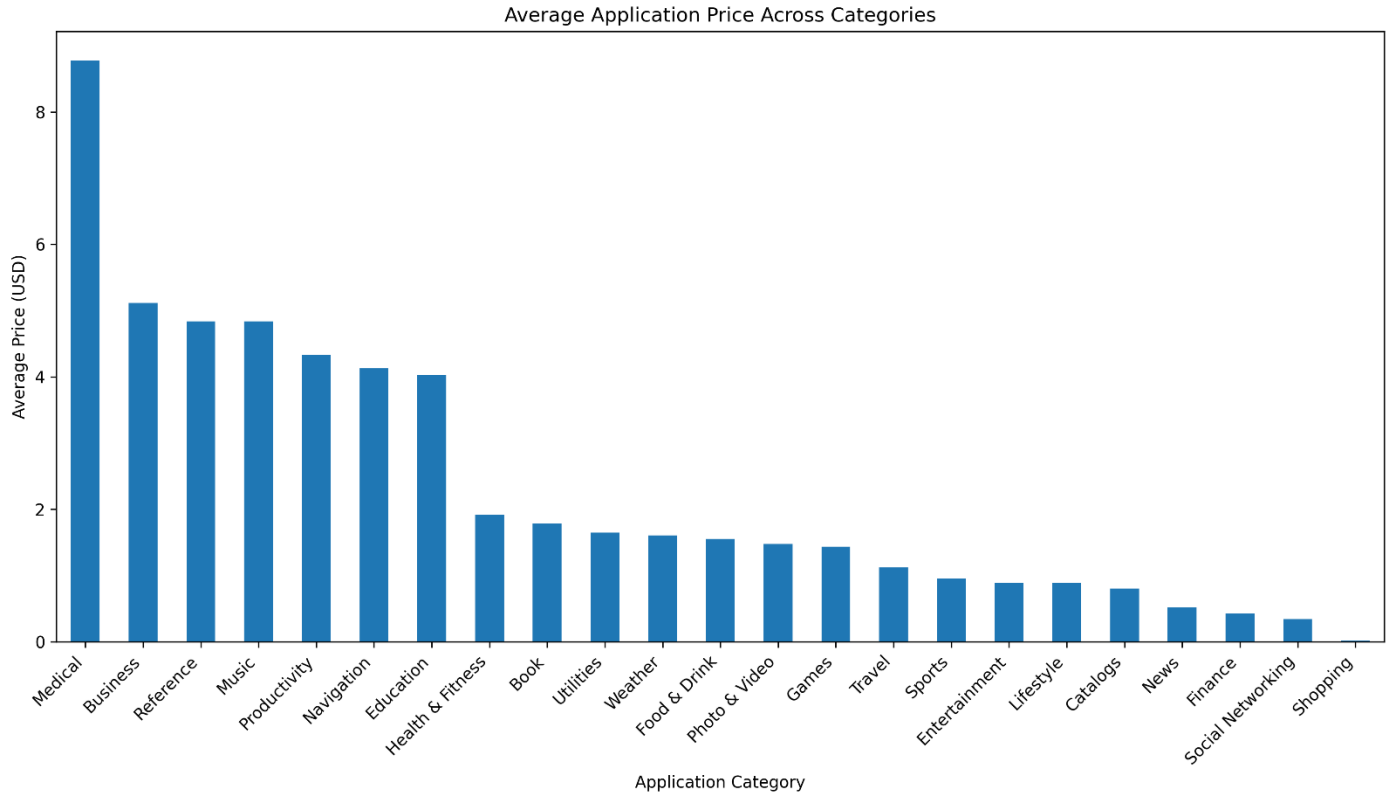
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- ## Insights:

- The graph highlights the **10 applications with the highest total rating counts** in the dataset.
- Applications at the top of the graph have received the **highest level of user engagement**.

- There is a noticeable difference in total rating counts among the selected top applications.
- The comparison shows that user engagement is **not evenly distributed** across applications.
- Overall, the graph identifies the applications that have attracted the **largest number of user ratings** in the Apple App Store dataset.

8. Application Pricing Across Category:



- **Interpretation:**

The graph represents the average price of applications across different categories in the Apple App Store dataset. The prices are measured in US dollars (USD) and show how the average application price differs from one category to another.

The Medical category has the highest average application price at approximately $8.78. It is followed by Business, with an average price of approximately $5.12, and Reference, with an average price of approximately $4.84. The Music category has an average price of approximately $4.84, while

Productivity has an average price of approximately $4.33.

The average price for Navigation applications is approximately $4.12, followed by Education at $4.03. Health & Fitness has an average price of approximately $1.92, while Book applications have an average price of approximately $1.79.

The categories Utilities ($1.65), Weather ($1.61), Food & Drink ($1.55), Photo & Video ($1.47), and Games ($1.43) fall within a similar lower-price range. Travel has an average price of approximately $1.12, while Sports has an average price of approximately $0.95.

At the lower end of the graph, Entertainment has an average price of approximately $0.89, Lifestyle approximately $0.89, Catalogs approximately $0.80, and News approximately $0.52. Finance has an average price of approximately $0.42, while Social Networking has an average price of approximately $0.34. The lowest average price is observed in the Shopping category, at approximately $0.02.

Overall, the graph presents the average application price for all 23 categories, ranging from approximately $8.78 in Medical to approximately $0.02 in Shopping.

- **Insights**
  - The graph compares the **average price of applications across different categories**.
  - Some categories have a **higher average application price**, while others have a comparatively lower average price.
  - The average price varies across categories, indicating differences in pricing patterns.
- The comparison helps identify categories where applications are generally priced higher or lower.

9. Summary of Key Metrics

Metric	Key Result
Total application	7,197
Largest application category	Games
Free vs paid application	Free application are the majority
User rating distribution	4.0-4.5
Higher user engagement	Facebook

9. Next Steps to Move From Descriptive → Predictive

descriptive analysis provides an initial undeThe rstanding of the Apple App Store dataset. The next step would be to use the identified patterns for deeper analysis and better decision-making.

- **Predictive Analysis:** Future analysis can be used to predict application performance based on available data such as ratings, price, and user engagement.
- **Application Performance:** Applications can be further compared using user ratings and total rating counts to understand their performance.
- **Pricing Analysis:** Further analysis can examine the relationship between application price and user ratings or user engagement.
- **Category Analysis:** Individual categories can be studied in greater detail to understand differences in application characteristics and performance.
- **Advanced Visualization:** Additional dashboards and visualizations can be developed to monitor

important application-level metrics more effectively.

These next steps can help extend the current analysis beyond basic descriptive findings and provide deeper insights into the Apple App Store dataset.

10.Reference

- **Kaggle — Apple App Store dataset(Kaggle — Mobile App Store (7200 apps)**
- **Python Pandas** (data handling/analysis)
- **Matplotlib** (graphs/visualization)
- **Microsoft Word** (report preparation)